

Notes: Markowitz (JF, 1952)

Portfolio Selection

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1 Big picture

- **Question.** How should an investor choose a portfolio once he has beliefs about the future performance of securities?
- **Stage of the problem.** Markowitz explicitly studies only the *second stage* of portfolio choice: given beliefs about returns, how should the investor allocate wealth across securities?
- **Core challenge.** A simple rule such as “maximize discounted expected return” cannot explain diversification. If that rule were correct, the investor would generally put all wealth in the single asset with the highest discounted expected payoff.
- **Main theoretical result.** Portfolio choice should be organized around two objects: **expected return** and **variance of return**. Efficient portfolios are those that minimize variance for a given expected return, or equivalently maximize expected return for a given variance.
- **Why this matters.** This paper provides the conceptual foundation of modern portfolio theory. The key idea is that risk is not only about the riskiness of each asset individually; it is also about *covariance* across assets.
- **Central insight about diversification.** Diversification is valuable not because the investor holds “many” securities mechanically, but because the investor combines securities whose returns are not too positively correlated.
- **Geometric contribution.** Markowitz shows that the portfolio problem can be visualized in mean-variance space. The attainable set of portfolios has an efficient boundary, and investors should choose on that boundary.
- **Economic takeaway.** Rational portfolio choice is a trade-off between mean and risk. The right diversification problem is therefore not “how many assets should I hold?” but “how do expected return, variance, and covariance jointly determine the efficient frontier?”

2 Model setup and key objects

2.1 Securities, returns, and portfolio weights

- There are N securities.
- Let X_i denote the share of wealth invested in security i .
- Markowitz imposes the constraints

$$X_i \geq 0 \quad \text{for all } i, \quad \sum_{i=1}^N X_i = 1,$$

so the baseline analysis rules out short sales.

- In the paper’s dynamic opening discussion, security i has anticipated one-dollar return r_{it} at date t , discounted by factor d_{it} .

- Markowitz then moves to a static representation in which security i delivers a random return R_i , and the portfolio return is

$$R = \sum_{i=1}^N X_i R_i.$$

2.2 Beliefs: means, variance, and covariance

- Let the expected return of security i be

$$\mu_i = \mathbb{E}[R_i].$$

- Let the covariance between securities i and j be

$$\sigma_{ij} = \text{Cov}(R_i, R_j).$$

- In particular, σ_{ii} is the variance of security i .
- These beliefs are treated as the investor's relevant inputs into the portfolio problem.

2.3 Portfolio mean and variance

The portfolio's expected return is

$$E = \mathbb{E}[R] = \sum_{i=1}^N X_i \mu_i,$$

and its variance is

$$V = \text{Var}(R) = \sum_{i=1}^N \sum_{j=1}^N X_i X_j \sigma_{ij}.$$

Interpretation. These two equations are the core of the paper. The first says portfolio return is a weighted average of individual expected returns. The second says portfolio risk depends on all pairwise covariances, not just on each asset's own variance.

2.4 Efficient portfolios

- A portfolio is **efficient** if no other feasible portfolio has:
 - lower variance for the same or higher expected return, or
 - higher expected return for the same or lower variance.
- Equivalently, efficient portfolios are those with:
 - minimum V for given E , or
 - maximum E for given V .

Interpretation. Mean and variance together define the opportunity set. The investor should never choose an interior dominated point when a better mean-variance combination exists on the boundary.

2.5 Scope and limitations of the paper

- The paper studies only the **choice** stage, not the formation of beliefs.
- The analysis assumes **static probability beliefs**. Markowitz notes that in a fuller treatment the probability distribution of yields should evolve over time.
- The paper presents the 3- and 4-security cases geometrically and does not give the full general analytic derivation for arbitrary N .
- The paper abstracts from market imperfections and from short sales.

3 Models and theoretical results (all equations + interpretation)

3.1 Why discounted expected return maximization fails

Markowitz first considers the rule that the investor should maximize discounted expected return. In the paper's notation, discounted anticipated return of the portfolio is

$$R = \sum_{t=1}^{\infty} \sum_{i=1}^N d_{it} r_{it} X_i.$$

If we define the discounted return of security i as

$$R_i = \sum_{t=1}^{\infty} d_{it} r_{it},$$

then portfolio return becomes

$$R = \sum_{i=1}^N X_i R_i.$$

Because R is a weighted average of the R_i 's and $X_i \geq 0$ with $\sum_i X_i = 1$, maximizing R means choosing

$$X_i = 1 \quad \text{for one asset with maximal } R_i.$$

Interpretation. The expected-return rule produces corner solutions. It gives no general reason to diversify. This is the paper's first big punchline.

3.2 Expected return-variance rule

Markowitz replaces the discounted-return rule with the **expected return-variance** rule. The investor views expected return as desirable and variance as undesirable.

For a portfolio with weights X_i ,

$$E = \sum_{i=1}^N X_i \mu_i, \quad V = \sum_{i=1}^N \sum_{j=1}^N X_i X_j \sigma_{ij}.$$

The investor should choose from the set of efficient (E, V) combinations.

Interpretation. The paper does *not* say that investors maximize expected return alone or minimize variance alone. The problem is inherently two-dimensional: one trades off mean against risk.

3.3 Statistical building blocks

For a random variable Y taking values $y_1, \dots, y_N$ with probabilities $p_1, \dots, p_N$,

$$\mathbb{E}[Y] = p_1 y_1 + p_2 y_2 + \dots + p_N y_N,$$

and

$$\text{Var}(Y) = p_1 (y_1 - \mathbb{E}[Y])^2 + \dots + p_N (y_N - \mathbb{E}[Y])^2.$$

For two random variables R_i and R_j ,

$$\sigma_{ij} = \text{Cov}(R_i, R_j) = \mathbb{E}[(R_i - \mathbb{E}[R_i])(R_j - \mathbb{E}[R_j])].$$

If

$$R = a_1 R_1 + a_2 R_2 + \dots + a_n R_n,$$

then

$$\mathbb{E}[R] = \sum_{i=1}^n a_i \mathbb{E}[R_i], \quad \text{Var}(R) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \sigma_{ij}.$$

Interpretation. These are the exact statistical facts that allow portfolio choice to be reduced to means and covariances.

3.4 Three-security case and geometry of the attainable set

With three securities and no short sales,

$$E = \sum_{i=1}^3 X_i \mu_i, \quad V = \sum_{i=1}^3 \sum_{j=1}^3 X_i X_j \sigma_{ij}, \quad X_1 + X_2 + X_3 = 1, \quad X_i \geq 0.$$

Using $X_3 = 1 - X_1 - X_2$, expected return can be written as

$$E = \mu_3 + X_1(\mu_1 - \mu_3) + X_2(\mu_2 - \mu_3),$$

while variance becomes a quadratic form in (X_1, X_2) .

The feasible portfolio set is the triangle in (X_1, X_2) space defined by

$$X_1 \geq 0, \quad X_2 \geq 0, \quad 1 - X_1 - X_2 \geq 0.$$

Interpretation. This is the basic picture: the investor chooses a point inside a feasible simplex, and every such point corresponds to a mean-variance pair.

3.5 Isomean lines, isovariance curves, and the critical line

In the three-security case:

- **Isomean curves** are straight parallel lines, because E is linear in the portfolio weights.
- **Isovariance curves** are concentric ellipses, because V is quadratic in the portfolio weights.

Let X denote the portfolio with minimum variance. For any given expected return E , the portfolio that minimizes variance lies where the isomean line is tangent to an isovariance ellipse. As E varies, these tangency points trace out the **critical line**.

Interpretation. The critical line is the locus of minimum-variance portfolios conditional on target expected return. The efficient set is part of this critical line together with boundary segments when the tangency path hits the feasible boundary.

3.6 Efficient set in the N-security case

Markowitz emphasizes that the same logic extends beyond three assets.

- In the 4-security case, the efficient set is again a sequence of connected line segments.
- More generally, in the N -security case, the efficient set starts at the portfolio of minimum variance and ends at the portfolio of maximum expected return.
- In (E, V) space, the efficient frontier is a sequence of connected parabola segments.

Interpretation. The familiar modern efficient frontier is already here conceptually. Markowitz's geometry shows that the efficient set is not arbitrary: it has a precise structure inherited from linear expected return and quadratic variance.

3.7 Diversification and covariance logic

The paper's main economic insight about diversification is that it depends on **covariance**, not just on the number of securities held.

- Holding many securities from the same industry may still leave the portfolio poorly diversified.
- Diversification is stronger when one combines securities whose returns are less positively correlated.
- Firms in different industries often have lower covariance than firms in the same industry, so cross-industry diversification is useful.

Markowitz also notes that if an investor diversifies between two portfolios with equal variance, the compound portfolio will typically have lower variance than either one, unless the returns are perfectly correlated.

Interpretation. This is why the paper is revolutionary. Risk is a property of the *joint distribution* of returns, not a simple sum of standalone asset risks.

3.8 Investment versus speculation

Near the end of the paper, Markowitz argues that the mean-variance principle is more plausible as a rule for **investment** than for **speculation**.

- If utility depends only on E and V , the investor will not accept an actuarially fair gamble.
- If utility also depends positively on the third moment M_3 , then some fair bets may be acceptable.

Interpretation. Mean-variance analysis is meant as a disciplined theory of prudent investing, not as a full theory of gambling preferences or higher-moment speculation.

3.9 Practical use of the mean-variance rule

Markowitz closes by explaining how the rule might be used in practice.

- One needs estimates of μ_i and σ_{ij} .
- Statistical methods can generate a tentative set of estimates.
- Judgment then revises those estimates to account for relevant information not captured by formal calculations.
- Given these beliefs, one computes the attainable efficient set, chooses the preferred (E, V) point, and then backs out the portfolio that implements it.

Interpretation. The paper is not only conceptual. It also proposes a concrete decision procedure: estimate beliefs, trace the efficient frontier, and then choose a portfolio on it.

4 Identification and logic (what makes the argument credible)

4.1 Why the old rule fails

- The paper does not reject expected-return maximization on psychological grounds.
- It rejects it by **logic**: the rule implies corner solutions and therefore cannot rationalize diversification.
- Since diversification is empirically common and intuitively sensible, a theory that never recommends it is inadequate.

4.2 What identifies the correct trade-off

- The expected return of a portfolio depends linearly on portfolio weights.
- The variance depends quadratically on weights through the covariance matrix.
- Once these two objects are accepted as the relevant ingredients, the efficient set follows from pure geometry.

Interpretation. The paper’s “identification” is conceptual rather than econometric. The structure of the problem comes from probability theory and budget constraints.

4.3 Why covariance is the decisive object

- A portfolio is not well diversified merely because it contains many assets.
- Diversification only reduces risk when one avoids assets whose returns move together.
- Covariance is therefore the relevant sufficient statistic for the *interaction* among securities.

Interpretation. The paper’s major conceptual move is to redefine risk at the portfolio level. The correct object is portfolio variance, and portfolio variance is built from covariances.

4.4 Main assumptions and limitations

- No short sales in the baseline setup.
- Static beliefs rather than a full dynamic stochastic process.
- The paper does not solve the general N -security case analytically in full detail.
- The first stage — forming beliefs from data and information — is left for future work.
- Higher moments matter for some speculative behavior, but the paper deliberately focuses on mean and variance.

4.5 Relation to the literature

- Markowitz explicitly frames his paper against the older discounted expected return tradition associated with investment value.
- He also notes a connection to Hicks’s discussion of anticipated returns with risk adjustments.
- The paper’s real legacy is that it establishes mean-variance efficiency as the modern language for portfolio choice.

5 Figures

There are no tables in the paper. The argument is carried almost entirely by the sequence of geometric figures.

5.1 Figure 1: Attainable and efficient (E,V) combinations

Read:

- Figure 1 introduces the most important object in the paper: the set of attainable mean-variance combinations.
- Only the boundary segment labeled efficient matters for choice; interior points are dominated.
- This is the conceptual birth of the efficient frontier.

5.2 Figure 2: Three-security geometry and the critical line

Read:

- The figure shows the feasible triangle in weight space, parallel isomean lines, and concentric isovariance ellipses.
- The critical line traces tangencies between isomean lines and isovariance curves.
- The efficient set is the relevant portion of this critical line together with a boundary segment.

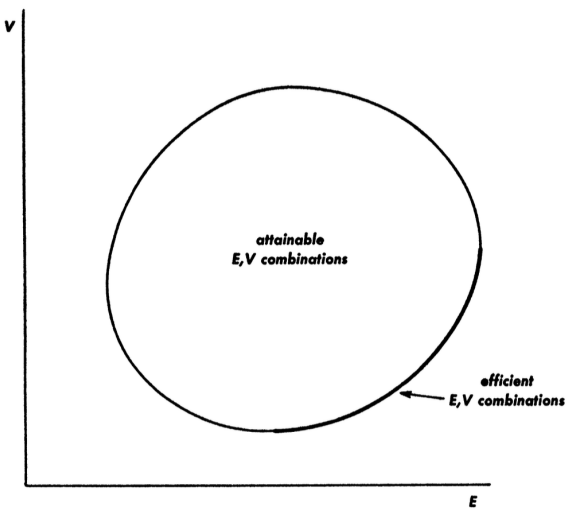


FIG. 1

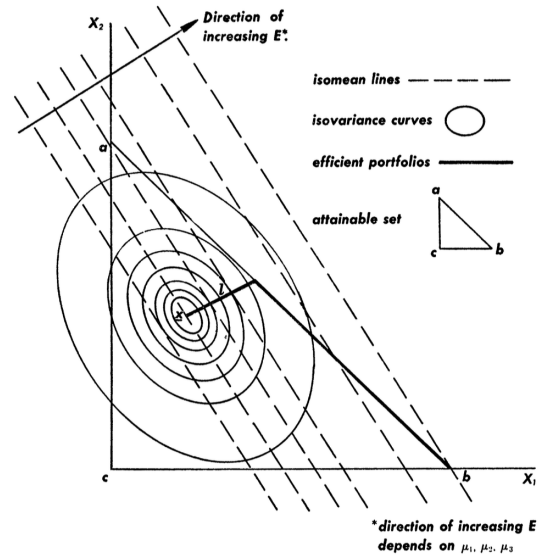


FIG. 2

5.3 Figures 3 and 4: Alternative cases for the efficient set

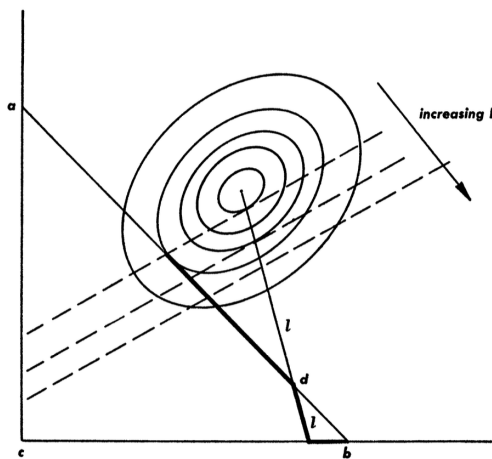


FIG. 3

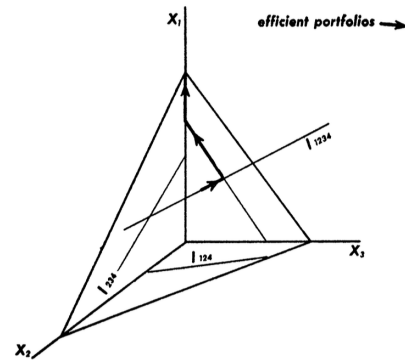


FIG. 4

Read:

- These figures show that the minimum-variance point can lie inside or outside the attainable set.
- Depending on that geometry, the efficient set may run partly along a boundary and partly along the critical line.
- This helps the reader see that efficiency is a constrained optimization object, not just a free tangency result.

5.4 Figures 5 and 6: Efficient portfolios and efficient (E,V) pairs

Read:

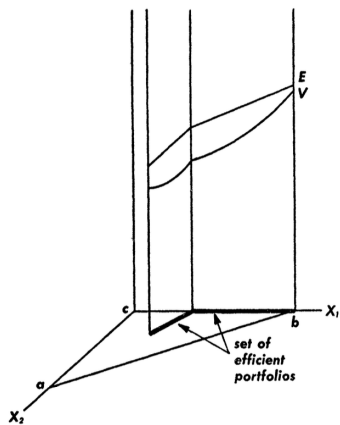


FIG. 5

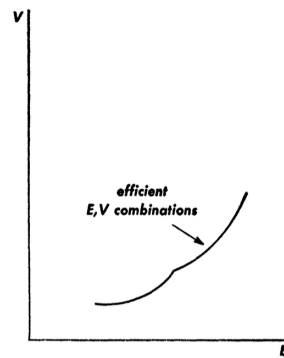


FIG. 6

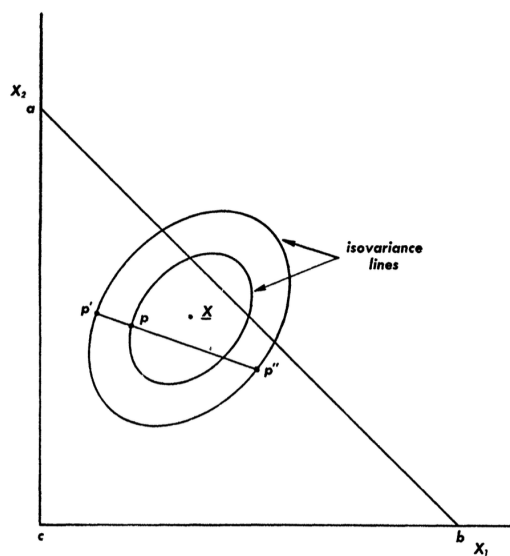


FIG. 7

- Figure 5 represents the efficient portfolio set in portfolio-weight space.
- Figure 6 maps that set into mean-variance space.
- Markowitz's point is that, for any number of securities, the efficient (E,V) frontier is built from connected parabola segments.

5.5 Figure 7: Diversification between portfolios

Read:

- The figure shows that diversification can also be understood as combining entire portfolios.
- If two portfolios have equal variance, their convex combination will typically have lower variance unless returns are perfectly correlated.
- This reinforces the paper's core message that covariance is the source of risk reduction.

6 Short summary

- **Method.** Markowitz analyzes portfolio choice using only beliefs about expected returns, variances, and covariances.
- **Core theoretical move.** He rejects discounted expected return maximization because it cannot explain diversification and replaces it with the expected return-variance rule.
- **Main result.** Efficient portfolios are those on the mean-variance frontier: minimum variance for a given expected return, or maximum expected return for a given variance.
- **Economic insight.** Diversification works through low covariance, not just through holding many assets.
- **Legacy.** The paper provides the fundamental logic behind modern portfolio theory and the efficient frontier.

7 Conclusion

7.1 Core conclusion

- The investor should not choose portfolios by maximizing expected return alone.
- Rational choice requires evaluating both expected return and variance.
- The correct portfolio problem is therefore a constrained mean-variance optimization problem.

7.2 Economic intuition

- Expected return rewards concentration in high-return assets.
- Variance penalizes concentration when returns co-move.
- Covariance makes diversification economically meaningful because combining imperfectly correlated assets reduces portfolio risk.

7.3 Takeaway

Markowitz's paper is short, but its logic is foundational. It shows that portfolio theory must be about *portfolios*, not isolated securities. The right primitive objects are expected return, variance, and covariance; the right solution concept is the efficient frontier; and the right understanding of diversification is that it is a covariance-management problem. These are the core ideas that still organize modern asset allocation.